Tutorial - 1

9/01/26

1. A smart farming system predicts whether a plant will grow well (1) or poorly (0) based on daily sunlight hours and water amount in liters

Dataset:

Sunlight Hours (x_1)	Water liters (x_2)	Bias (x_3)	Target (Good Growth = 1 / Poor = 0)
3	1	1	0
8	4	1	1
5	2	1	0
10	5	1	1
7	3	1	1

Sol:-

Inputs (x_1 & x_2) output (Target)

- mean / Average = central tendency of the data
- varience / standard deviation.
- mean of $x_1 = \frac{3+8+5+10+1}{5} = \frac{33}{5} = 6.6 \Rightarrow l_1 x_1$
- mean of $x_2 = \frac{1+4+2+5+3}{5} = \frac{15}{5} = 3 \Rightarrow l_1 x_2$
- Variance, $\boxed{(\sigma^2) = \frac{1}{N} \sum_i (x_i - \mu)^2}$

$N \rightarrow$ no. of data

$x_i \rightarrow$ input item

$$= \frac{(3-6.6)^2 + (3-6.6)^2 + (5-6.6)^2 + (10-6.6)^2 + (7-6.6)^2}{5}$$

$$= \frac{146}{25} \quad 5.84$$

$$= \frac{(1-3)^2 + (4-3)^2 + (2-3)^2 + (5-3)^2 + (3-3)^2}{5}$$

$$= 2$$

= standard Deviation $= \sqrt{\sigma^2_{x_1}} = \sqrt{(5.8)} = 2.40$

$= \sqrt{\sigma^2_{x_2}} = \sqrt{2} = 1.414$

ML model → perception model

=> simplest neutral networks (N.N) model

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simple decision making model that can answer yes/NO

=> preception:

1) Inputs :- x_1 and x_2

2) weights associated with inputs : w_1 and w_2

3) Bias → model flexiability (b)

4) weighted sum : $z = w_1 x_1 + w_2 x_2 + b$

$$= \boxed{\sum_i w_i x_i + b}$$

5) Activation function $\hat{y} = \begin{cases} 1; & \text{if } z \geq 0 \\ 0; & \text{if } z < 0 \end{cases}$

(step function)

6) updated weights

$$w_i^{new} = \boxed{w_i^{old} + \eta(y-\hat{y}) \times x_i}$$

$\eta \rightarrow$ learning Rate

$y \rightarrow$ Actual output

$\hat{y} \rightarrow$ predicted Target

Prediction

1) Initial :-

Inputs : $x_1 = 3$; $x_2 = 1$; $x_3 = 1$

$Y = 0$

let $w_1 = 0$

$w_2 = 0$

$\eta = 1$

$z = ?$

$$z = w_1 x_1 + w_2 x_2 + b$$
$$= 0 \times 3 + 0 \times 1 + 1$$
$$= 1$$

$z = 1$; $\hat{y} = 1$

$$w_1^{new} = 0 + 1\,(0-1) \times 3 = -3$$

$$w_2^{new} = 0 + 1\,(0-1) \times 1 = -1$$

Now, $w_1 = -3$

$w_2 = -1$

$\eta = 1$

$b = 1$

Inputs: $x_1 = 8$; $x_2 = 4$; $b = 1$

$z = w_1 x_1 + w_2 x_2 + b$

$= -3 \times 8 + -1 \times 4 + 1$

$= -24 - 4 + 1$ $\qquad y = 1$

$= -27;$

$x = -27; \quad \hat{y} = 0$

$w_1^{new} = -3 + 1(1-0) \times 8 = -2 \times 8 = 5$

$w_2^{new} = -1 + 1(1-0) \cdot 4 = 0 = 3$

Now, $w_1 = 5$ $\qquad$ Inputs: $x_1 = 5; x_2 = 2; b = 1$

$w_2 = 3$ $\qquad y = 0$

$\eta = 1$ $\qquad z = 5 \times 5 + 3 \times 2 + 1$

$b = 1$ $\qquad = 25 + 6 + 1$

$z = 32 \quad : \quad \hat{y} = 1$

$w_1^{new} = 5 + 1(0-1) \times 5 = 0$

$w_2^{new} = 3 + 1(0-1) \times 2 = 1$